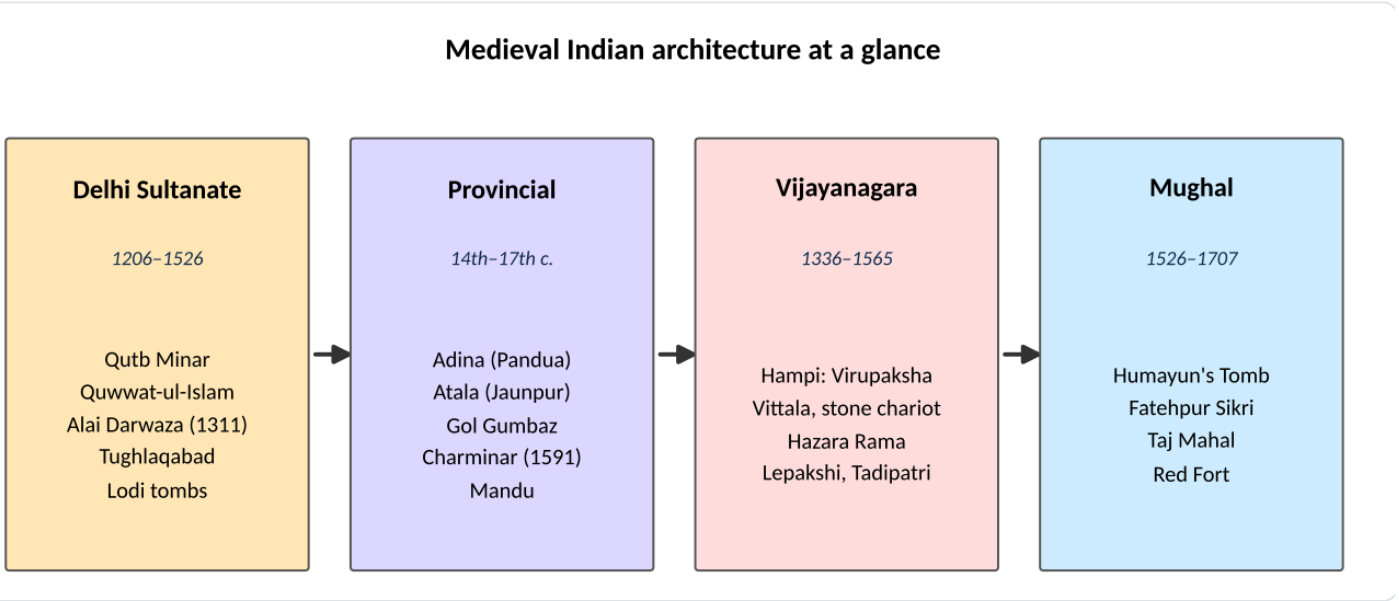
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HISTORY · Medieval art, architecture, language and literature

UNDERSTAND IT FIRST

What is it? Rulers, nobles and merchants of the medieval age (c. 1200–1750) paid for buildings, paintings, music and books. This **patronage** shows us **what each age valued**. Medieval culture is a **meeting of traditions**: Turko-Persian ideas met Indian ones and created a new **Indo-Islamic** style, while in the south the **Vijayanagara** kings carried on and grew the Dravidian temple style. Learn it as **Architecture → Painting → Music and dance → Language and literature**.



Medieval architecture

1. What is Indo-Islamic architecture?

Feature	Meaning
Arch, dome, vault (arcuate)	Weight is carried by **curved arches**; the older Indian style was **trabeate** (beam and pillar). **Voussoirs** are the wedge stones of an arch.
Minaret	A tall tower beside a mosque from which the call to prayer (**azan**) is given.
Lime mortar	Mortar of lime and sand made large domes and arches strong.
Calligraphy, arabesque, geometry	Walls carry **Quranic verses** and **floral and geometric patterns** instead of images of living beings. **Jali** = pierced stone screen; later **pietra dura** = inlay of coloured stones.
Indian contribution	Indian **craftsmen**, lotus, bell and chain motifs, **brackets** and **chhatris** joined the foreign forms.

2. Delhi Sultanate (1206–1526)

Dynasty	Buildings and facts

Slave dynasty	**Quwwat-ul-Islam** mosque (Delhi, Aibak, using material from temples); **Adhai-din-ka-Jhonpra** (Ajmer, Aibak); **Qutb Minar** (begun by **Aibak**, completed by **Iltutmish**; later floors repaired by **Firoz Shah Tughlaq**); **Sultan Ghari** (Iltutmish's son's tomb, c. 1231, the first Islamic tomb in India).
Khalji	**Alai Darwaza** (1311, Alauddin Khalji): true **arch and dome** built on proper principles, **horseshoe arches**, red sandstone with white marble inlay; the city of **Siri**; **Jamaat Khana mosque**.
Tughlaq	Plain, strong style with **sloping walls**: **Tughlaqabad** (Ghiyasuddin Tughlaq); his **tomb**; **Jahanpanah** and **Adilabad** (Muhammad bin Tughlaq); **Firozabad** and **Firoz Shah Kotla**, **Hauz Khas**, many canals and gardens (Firoz Shah).
Sayyid and Lodi	**Octagonal tombs** and **square tombs**; the **double dome** appears; tombs set in gardens (**Lodi Garden**, tombs of Sikandar Lodi, Muhammad Shah); the model for later Mughal **garden tombs**.

3. Provincial styles

Region	Notable buildings
Jaunpur (Sharqi)	**Atala Masjid** with a huge central gateway; **Jama Masjid**.
Bengal	**Brick** and terracotta, curved roofs: **Adina Mosque** (Pandua, Sikandar Shah, 1370s, once the largest in the region); **Eklakhi tomb**.
Gujarat	Hindu-Jain carving with Islamic form: **Ahmedabad Jama Masjid**, **Sidi Saiyyed mosque's jali** (famous tree-of-life screen), step-wells.
Malwa	**Mandu**: **Hoshang Shah's tomb** (first marble tomb in India), **Jahaz Mahal**, **Hindola Mahal**, **Baz Bahadur's palace**.
Kashmir	Wooden mosques with Buddhist-style roofs, e.g. **Shah Hamadan mosque**, **Zain-ul-Abidin** era.
Deccan (Bahmani, Adil Shahi, Qutb Shahi)	**Gulbarga** and **Bidar** forts; **Mahmud Gawan's Madrasa** (Bidar); **Gol Gumbaz** (Bijapur, Muhammad Adil Shah, 1656, one of the world's largest domes, a 'whispering gallery'); **Ibrahim Rauza**; **Charminar** (Hyderabad, **Muhammad Quli Qutb Shah**, 1591); **Golconda fort**; **Mecca Masjid**.

4. Vijayanagara and temple architecture

- **Vijayanagara style** (capital **Hampi**, 1336–1565): tall **gopurams** (gateway towers), **kalyana mandapas** (wedding halls), **hundred-pillared halls**, **carved musical pillars**, the **stone chariot**. Main sites: **Virupaksha Temple**, **Vittala Temple**, **Hazara Rama Temple**, **Lotus Mahal**, **Elephant stables**, **Krishna temple** (built by **Krishnadevaraya**). Domingo Paes and Nuniz praised the city.
- **Andhra region: Lepakshi** (Veerabhadra temple, 16th c., built under Virupanna and Viranna; giant **Nandi**, **hanging pillar**, murals), **Tadipatri** (Chintalarayaswami and Bugga Ramalingeswara temples), **Srikalahasti** (gopuram added by Krishnadevaraya), **Tirumala** (gopuram and mandapas), **Ahobilam**, **Srisailam**, **Kalyanadurgam** and **Penukonda**: spread across Rayalaseema, Coastal Andhra and Telangana.

- **Other medieval styles: Chola** (**Brihadisvara temple**, Thanjavur, Rajaraja I, c. 1010); **Hoysala** (**Belur**, **Halebidu**: star-shaped base, soapstone carving); **Rajput** (**Vijay Stambh**, Chittor, Rana Kumbha; forts of **Mehrangarh**, **Amber**, **Kumbhalgarh**); **Sikh** (**Harmandir Sahib**, Amritsar).

Mughal architecture was covered on Day 9 (Humayun's Tomb, Fatehpur Sikri, Taj Mahal, Red Fort); remember its steps: **Humayun's Tomb → Fatehpur Sikri → Itimad-ud-Daula → Taj Mahal**.

5. Painting

School	Features
Persian miniature, Sultanate	Illustrated manuscripts; Persian influence; Jain and **Apabhramsha** manuscript painting continued in Gujarat and Rajasthan.
Mughal	Persian + Indian: **Humayun** brought Mir Sayyid Ali and Abdus Samad; **Akbar's** workshop (**Hamzanama**, **Razmnama**); **Jahangir's** natural-history and portrait painting (Mansur, Abul Hasan); fine lines, realism, shading.
Rajput	Hindu themes: **Ramayana**, **Gita Govinda**, **Ragamala**, Krishna; bright colours. Schools: **Mewar, Bundi, Kota, Jaipur, Kishangarh** (Nihal Chand's **Bani Thani**), Marwar.
Pahari	Himalayan foothills: **Basohli** (bold), **Guler, Kangra** (soft, lyrical, 18th c.), Garhwal.
Deccani	**Ahmednagar, Bijapur, Golconda**: rich colour, Persian, Turkish and local mix; **Ragamala** paintings.

6. Music and dance

- **Amir Khusrau** (Delhi, Alauddin Khalji's time): poet and musician, traditionally credited with mixing Persian and Indian music, **qawwali**, **khayal**, and the **sitar** and **tabla**; he wrote in **Hindavi**. **Sarangadeva's** *Sangita Ratnakara* (13th c.) is a key text.
- **Hindustani: Tansen** (Akbar's court, **Dhrupad**, ragas like **Miyan ki Malhar**), **Man Singh Tomar** of Gwalior (patron of Dhrupad), **Baiju Bawra**. **Karnatak: Purandaradasa** (15th–16th c., 'father of Karnatak music'), **Annamacharya** and the **Vijayanagara** court; later the Trinity (Tyagaraja, Shyama Shastri, Muthuswami Dikshitar) in the 18th–19th c.
- **Dance: Kathak** (north, temple storytelling that moved to Mughal courts), **Kuchipudi** (Andhra, developed by **Siddhendra Yogi**, 15th c., Kuchipudi village in Krishna district), **Bhagavata Mela** (Tamil Nadu), **Yakshagana** (Karnataka). **Bhakti** and **Sufi** devotional music (**kirtan**, **qawwali**, **abhangs**) spread.

7. Language and literature

Persian became the **language of the court and administration**; **Arabic** for religion and learning; **Sanskrit** continued for scholarship; and **regional languages** grew through **Bhakti** poets who spoke to the people. **Urdu** (camp language of Persian, Arabic, Turkish and Hindi words) and **Hindavi/Hindi** developed. **Translations** (Mahabharata as **Razmnama**, Ramayana, Atharva Veda, Panchatantra as **Anwar-i-Suhaili**) were made under Akbar.

Language / region	Authors and works
Persian and Arabic (histories and travellers)	**Al-Biruni** (*Kitab-ul-Hind*, with Mahmud of Ghazni, c. 1030); **Minhaj-us-Siraj** (*Tabaqat-i-Nasiri*); **Amir Khusrau** (*Khazain-ul-Futuh*, *Tughlaq Nama*, *Tuti-i-Hind* 'parrot of India'); **Ziauddin Barani** (*Tarikh-i-Firoz Shahi*); **Isami** (*Futuh-us-Salatin*); **Ibn Battuta** (*Rihla*, Muhammad bin Tughlaq's time); Vijayanagara visitors **Abdur Razzaq**, **Nicolo Conti**, **Domingo Paes**, **Nuniz**.
Sanskrit	**Kalhana's** *Rajatarangini* (1148, Kashmir; the best early history in Sanskrit); **Jayadeva's** *Gita Govinda* (Bengal, 12th c.); **Vidyaranya** and **Sayana** (Vijayanagara, Veda commentary); **Gangadevi's** *Madhura Vijayam*; Krishnadevaraya's Sanskrit works.
Hindi, Urdu, Punjabi	**Khusrau** (Hindavi riddles), **Kabir** (dohas), **Malik Muhammad Jayasi** (*Padmavat*, 1540, Awadhi), **Tulsidas** (*Ramcharitmanas*), **Surdas** (*Sur Sagar*), **Mirabai**, **Rahim**, **Bihari**; **Guru Nanak** and the **Adi Granth** (Punjabi, Gurmukhi script).
Other regional	**Bengali**: Chandidas, Krittivasa (Ramayana); **Assamese**: Sankaradeva; **Marathi**: Dnyaneshwar (*Dnyaneshwari*), Eknath, Tukaram; **Gujarati**: Narsinh Mehta; **Tamil**: Kamban (*Ramavataram*); **Malayalam**: Ezhuthachan; **Kannada**: Purandaradasa, Kanakadasa.

8. Telugu literature

Period / poet	Work and facts
Nannaya (11th c.)	**Adikavi**; began the **Andhra Mahabharatam** under **Rajaraja Narendra** (Eastern Chalukya, Rajahmundry); wrote the first two and a half parvas.
Kavitrayam (Trinity)	**Nannaya**, **Tikkana** (13th c., Kakatiya period, Nellore, *Nirvachanottara Ramayanam*, completed Mahabharata from the 4th to 18th parvas), **Errana** (14th c., finished the Aranya parva; *Harivamsam*).
Palkuriki Somanatha, **Srinatha**	Somanatha: *Basava Purana*, in **Dwipada** metre; **Srinatha** (15th c., 'Kavi Sarvabhauma'): *Srungara Naishadham*, *Kasi Khandam*, *Bhimakhandam*; his **Chatuvulu**.
Potana (15th c.)	**Andhra Maha Bhagavatam**; refused to dedicate it to a king, dedicating it to **Sri Rama**; from Bammera (Warangal region).
Krishnadevaraya (1509–29) and the **Ashtadiggajas**	Called **Andhra Bhoja**, **Telugu Vallabha**; wrote **Amuktamalyada** (the story of Andal / Godadevi) in Telugu. Court poets: **Allasani Peddana** (*Manucharitra*, **Andhra Kavita Pitamaha**), **Nandi Timmana** (*Parijatapaharanam*), **Madayyagari Mallana**, **Dhurjati** (*Srikalahasti Mahatmyam*), **Ayyalaraju Ramabhadrudu**, **Pingali Surana** (*Raghava Pandaviyam*), **Bhattumurti**, **Tenali Ramakrishna** (*Panduranga Mahatmyam*).
Annamacharya (1408–1503)	**Pada Kavita Pitamaha**; composed about 32,000 **sankirtanas** to Venkateswara at **Tirumala**; copper-plate manuscripts kept at Tirumala.

Later period	**Kancherla Gopanna (Ramadasu)** (Bhadrachalam temple, *Dasarathi Satakam*); **Vemana** (poet of simple moral verse); Nayaka courts of **Thanjavur** and **Madurai** (**Raghunatha Nayaka**, **Madhuravani**) kept Telugu literature alive after Vijayanagara.

EXAM TRAP

Alai Darwaza 1311 (first true dome); **Hoshang Shah's tomb** = first marble tomb; **Gol Gumbaz** (Bijapur 1656); **Charminar 1591** (Quli Qutb Shah); **Lepakshi** (Virupanna, Viranna); **Rajatarangini** (Kalhana, 1148); **Kitab-ul-Hind** (Al-Biruni); **Tarikh-i-Firoz Shahi** (Barani); **Padmavat** (Jayasi); **Amuktamalyada** (Krishnadevaraya); **Manucharitra** (Peddana); **Bhagavatam** (Potana); **Annamacharya**: Pada Kavita Pitamaha; **Kuchipudi**: Siddhendra Yogi; **Purandaradasa**: father of Karnatak music.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about Delhi Sultanate architecture:

1. The Alai Darwaza was built during the reign of Alauddin Khalji.

2. The Qutb Minar was begun by Aibak and completed by Iltutmish.

3. The double dome appeared in the tombs of the Lodi period.

4. Tughlaqabad was built by Muhammad bin Tughlaq.

Which of the above are correct?

A. 1 and 4 only

B. 2 and 3 only

C. 1, 2 and 3 only

D. 1, 2, 3 and 4

Answer: C. Tughlaqabad was built by Ghiyasuddin Tughlaq, the founder, not Muhammad bin Tughlaq.

PRACTICE

Match List-I (work) with List-II (author):

List-I (work)	List-II (author)
A. Rajatarangini	1. Kalhana
B. Kitab-ul-Hind	2. Al-Biruni
C. Padmavat	3. Malik Muhammad Jayasi
D. Tarikh-i-Firoz Shahi	4. Ziauddin Barani

(1) A → 2 B → 1 C → 3 D → 4

(2) A → 1 B → 2 C → 4 D → 3

(3) A → 1 B → 2 C → 3 D → 4

(4) A → 3 B → 2 C → 1 D → 4

Answer: (3) Rajatarangini: Kalhana; Kitab-ul-Hind: Al-Biruni; Padmavat: Jayasi; Tarikh-i-Firoz Shahi: Barani.

PRACTICE

Which one of the following statements is NOT correct?

A. The Vijayanagara kings built the stone chariot at the Vittala Temple, Hampi.

B. Amir Khusrau is traditionally credited with qawwali and with writing in Hindavi.

C. The Charminar at Hyderabad was built by Muhammad Quli Qutb Shah in 1591.

D. The Gol Gumbaz at Bijapur was built by the Bahmani sultans of Gulbarga.

Answer: D. Gol Gumbaz (1656) is the tomb of Muhammad Adil Shah of the Adil Shahi dynasty of Bijapur.

PRACTICE

Assertion (A): Indo-Islamic buildings are decorated with calligraphy, arabesque and geometric patterns rather than images of living beings.
Reason (R): Islamic religious tradition discourages depicting living beings in places of worship.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. Because figures were avoided, artists used writing and patterns.

PRACTICE

A Portuguese traveller visited Vijayanagara during the reign of Krishnadevaraya and described the markets and festivals of Hampi. Who was he?

A. Domingo Paes

B. Ibn Battuta

C. Francois Bernier

D. Jean-Baptiste Tavernier

Answer: A. Domingo Paes (c. 1520–22) described Vijayanagara; Nuniz wrote later. Bernier and Tavernier were French; Ibn Battuta was a Moroccan.

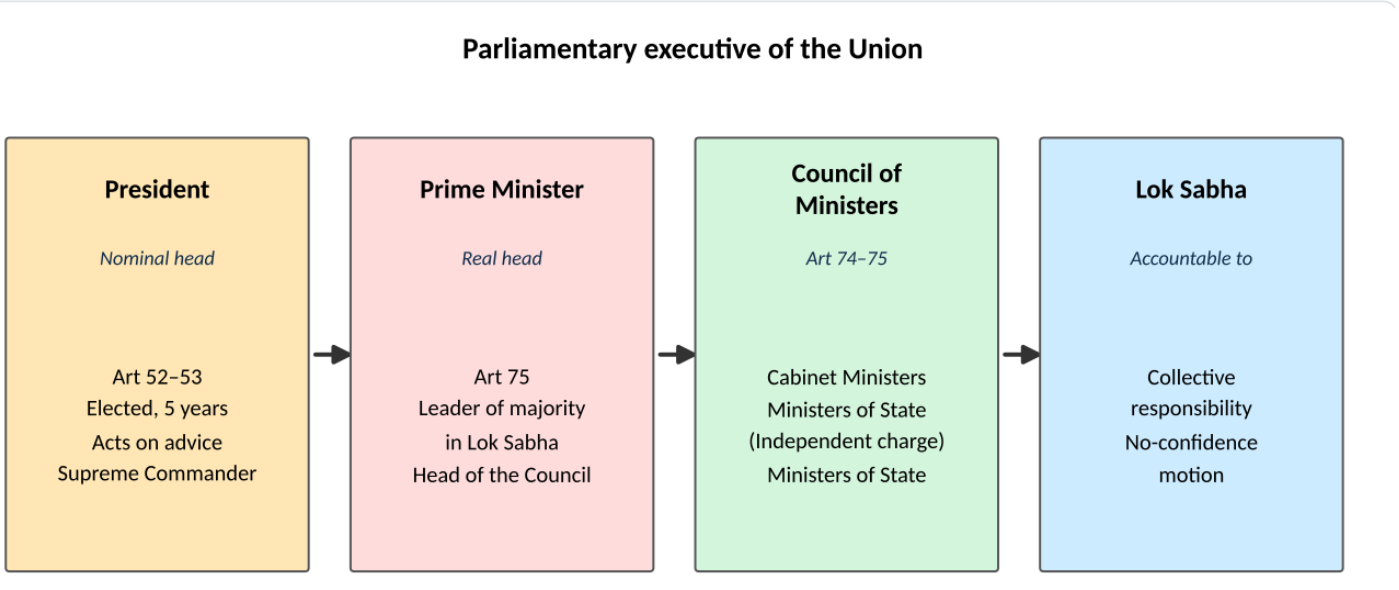
RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the features of Indo-Islamic architecture and the key buildings of each dynasty. 2. Write the painting schools and music facts. 3. Write the language and literature table and the Telugu poets with their works.

POLITY · Parliamentary executive: President, Prime Minister, Council of Ministers

UNDERSTAND IT FIRST

What is it? In a **parliamentary system** (the **Westminster model**) the **head of State** (President) is a **nominal** head, and the **real power** lies with the **Prime Minister and the Council of Ministers**, who are chosen from the legislature and must keep its **majority support**. In a **presidential system** (USA) the President is both head of State and real executive and is **not** answerable to the legislature in the same way. India chose the parliamentary system for **responsibility** and for **harmony** between legislature and executive.



The Union executive

1. The President

Point	Details
Position	**Art 52**: there shall be a President; **Art 53**: executive power of the Union is vested in the President, who uses it directly or through officers under the Constitution.
Qualifications (Art 58)	Citizen of India, **35 years** old, qualified to be a **Lok Sabha member**, holds **no office of profit**. Nomination needs **50 proposers and 50 seconders**.
Election (Art 54–55)	By an **electoral college**: **elected members** of both Houses of Parliament and **elected members** of the State Legislative Assemblies (including **Delhi and Puducherry**). **Nominated members** and members of **Legislative Councils do not vote**. Method: **proportional representation by single transferable vote**, **secret ballot**. Votes are weighted so that States and the Union carry equal weight.
Term (Art 56)	**5 years**, may be re-elected; resigns to the **Vice-President**; continues until the successor takes office. A vacancy must be filled by election **within 6 months**; the **Vice-President acts** meanwhile (Art 65).
Impeachment (Art 61)	Only for **violation of the Constitution**; either House can start; **14 days' notice** signed by at least **1/4** of its members; passed by **2/3 of the total membership**; the other House investigates, and also passes by 2/3. Nominated members **can** vote in this process.

Powers of the President

Type	Powers
Executive	Appoints the **PM**, ministers, **Governors**, judges, Attorney General, CAG, Chief Election Commissioner; **Supreme Commander** of the armed forces.
Legislative	**Summons and prorogues** Parliament, **dissolves the Lok Sabha**, addresses it, **nominates 12 to the Rajya Sabha** and 2 Anglo-Indians earlier; **ordinances** (Art 123, when Parliament is not in session); gives **assent** to Bills.
Financial	Money Bills need his **prior recommendation**; places the **Budget** before Parliament; controls the **Contingency Fund**; appoints the **Finance Commission** every 5 years.
Judicial	**Pardon** (Art 72: pardon, reprieve, respite, remission, commutation, including **death sentences**); seeks the Supreme Court's advice (**Art 143**).
Emergency	**National Emergency** (Art 352), **President's Rule** (Art 356), **Financial Emergency** (Art 360).
Veto (Art 111)	**Absolute** (withhold assent; only on private members' and government Bills after a Cabinet resigns), **suspensive** (return a Bill once, Parliament may pass it again), **pocket** (take no decision, no time limit). **No veto on Money Bills** returned, and a **Constitution amendment Bill must be signed**.

2. The Vice-President (Art 63–71)

Elected by an electoral college of **all members of both Houses of Parliament** (including nominated members, but **not** State legislators) by **single transferable vote**; **35 years**; **5-year term**; ex-officio **Chairman of the Rajya Sabha** (no vote except to break a tie); acts as President during a vacancy (up to 6 months).

3. The Prime Minister and the Council of Ministers

Provision	Details
Art 74	A **Council of Ministers headed by the PM** shall **aid and advise** the President, who **shall act** on that advice (**42nd Amendment**, 1976); the **44th Amendment** (1978) lets the President **ask the Council to reconsider once**, but he must act on the advice given **after reconsideration**. Courts cannot ask what advice was given.
Art 75(1)	The **PM is appointed by the President**; other ministers are appointed **on the PM's advice**. By convention the President appoints the **leader of the majority party or coalition** in the Lok Sabha. The PM may be a member of **either House**.
Art 75(1A), (1B)	**91st Amendment (2003):** the Council cannot exceed **15% of the Lok Sabha's strength** (in a State, 15% of the Assembly, minimum 12); a member **disqualified for defection** cannot be a minister.
Art 75(2)	Ministers hold office **during the pleasure of the President** (**individual responsibility**).
Art 75(3)	The Council is **collectively responsible to the Lok Sabha**: if it loses **confidence**, **all** must resign. A minister who disagrees must **resign or support**.
Art 75(5)	A minister who is **not a member of Parliament** must become one **within 6 months**, or he ceases to be a minister.
Art 77, 78	Business is conducted in the **President's name**; the **PM must inform the President** of all decisions on administration and legislation and give any information he asks for (**Art 78**).

Kinds of ministers

Rank	Role
Cabinet Ministers	Heads of major Ministries (Home, Defence, Finance, External Affairs); the **Cabinet** is the **real decision-making** body, about 25 members, meets weekly. The word 'Cabinet' entered the Constitution only through the **44th Amendment** (Art 352).
Ministers of State (Independent charge)	Run a small Ministry on their own; attend Cabinet when their subject is discussed.
Ministers of State	Assist Cabinet Ministers; do not attend Cabinet unless invited. (**Deputy Ministers**, once a rank, are now not appointed.)
Extra-constitutional bodies	**Cabinet Committees**, the **Cabinet Secretariat** headed by the **Cabinet Secretary**, and the informal **'Kitchen Cabinet'** of the PM's close advisers.

- **The Prime Minister: 'keystone of the Cabinet arch'** (Nehru's words for the office), **'first among equals'**. Roles: chairs the Cabinet, links the **President** and the Cabinet, leads the **Lok Sabha**, **allocates portfolios**, can **recommend dissolution**, heads the **Planning body (NITI Aayog)** and the **National Development Council**. If the PM resigns or dies, the **whole Council** goes.
- **Important cases: Ram Jawaya Kapur (1955)**: the President is a constitutional head and real power lies with the Council; **Shamsher Singh (1974)**: the President and Governors act on ministers' advice except in a few special cases; **U.N. Rao (1971)**: the Council continues even after the Lok Sabha is dissolved until a new House meets.
- **Attorney General** (Art 76): the **first law officer**; appointed by the President; qualified as a Supreme Court judge; can speak in Parliament but cannot vote. **Comparison:** the President's Art 74 position is like the **King in Britain**, but India is a **republic** (elected head) and has a **written Constitution**.

EXAM TRAP

Art 52–53: President; **35 years**; **electoral college**: elected MPs + elected MLAs (incl. Delhi, Puducherry), **STV, secret ballot**; **nominated MPs and MLCs do not vote**; term **5 yrs**; **impeachment Art 61**: 14 days, 1/4 notice, 2/3 of total; **Art 72** pardon; **Art 123** ordinance; **Art 74**: advice binding (**42nd**), reconsideration once (**44th**); **Art 75**: PM appointed by President, **15%** cap (**91st**), collective responsibility to **LS**, **6 months** rule; **VP**: elected by MPs only.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about the election of the President of India:
1. Elected members of the Legislative Assemblies of Delhi and Puducherry take part in it.
2. Nominated members of Parliament take part in it.
3. It is held by the system of proportional representation by means of the single transferable vote.
4. Members of State Legislative Councils take part in it.
Which of the above are correct?

A. 1 and 3 only

B. 1, 2 and 3 only

C. 2 and 4 only

D. 1, 3 and 4 only

Answer: A. Nominated MPs and Council members do not vote in the Presidential election.

PRACTICE

Match List-I (Article) with List-II (subject):

List-I (Article)	List-II (subject)
A. Article 61	1. Attorney General of India
B. Article 72	2. Ordinance-making power
C. Article 76	3. Impeachment of the President
D. Article 123	4. Power to grant pardons

(1) A → 3 B → 1 C → 4 D → 2

(2) A → 1 B → 3 C → 4 D → 2

(3) A → 3 B → 4 C → 1 D → 2

(4) A → 4 B → 3 C → 1 D → 2

Answer: (3) 61: impeachment; 72: pardon; 76: Attorney General; 123: ordinances.

PRACTICE

Which one of the following statements is NOT correct?

A. The 91st Amendment limits the Council of Ministers to 15% of the Lok Sabha's strength.

B. A person who is not a member of Parliament can be made a minister but must become a member within six months.

C. The Council of Ministers is collectively responsible to the Rajya Sabha.

D. The President may ask the Council of Ministers to reconsider its advice once.

Answer: C. Collective responsibility is to the Lok Sabha only (Art 75(3)).

PRACTICE

Assertion (A): The President must act on the advice of the Council of Ministers once it has been reconsidered.
Reason (R): The 44th Amendment to Article 74 allowed the President to send the advice back once, but made the advice given after reconsideration binding.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. R states exactly the rule that makes A true.

PRACTICE

A person who is not a member of either House of Parliament is sworn in as a Union Minister on 1 March. By which date must he become a member of either House to continue as a minister?

A. 1 September of the same year

B. 1 June of the same year

C. 1 March of the next year

D. There is no time limit

Answer: A. Art 75(5): six months from taking the oath.

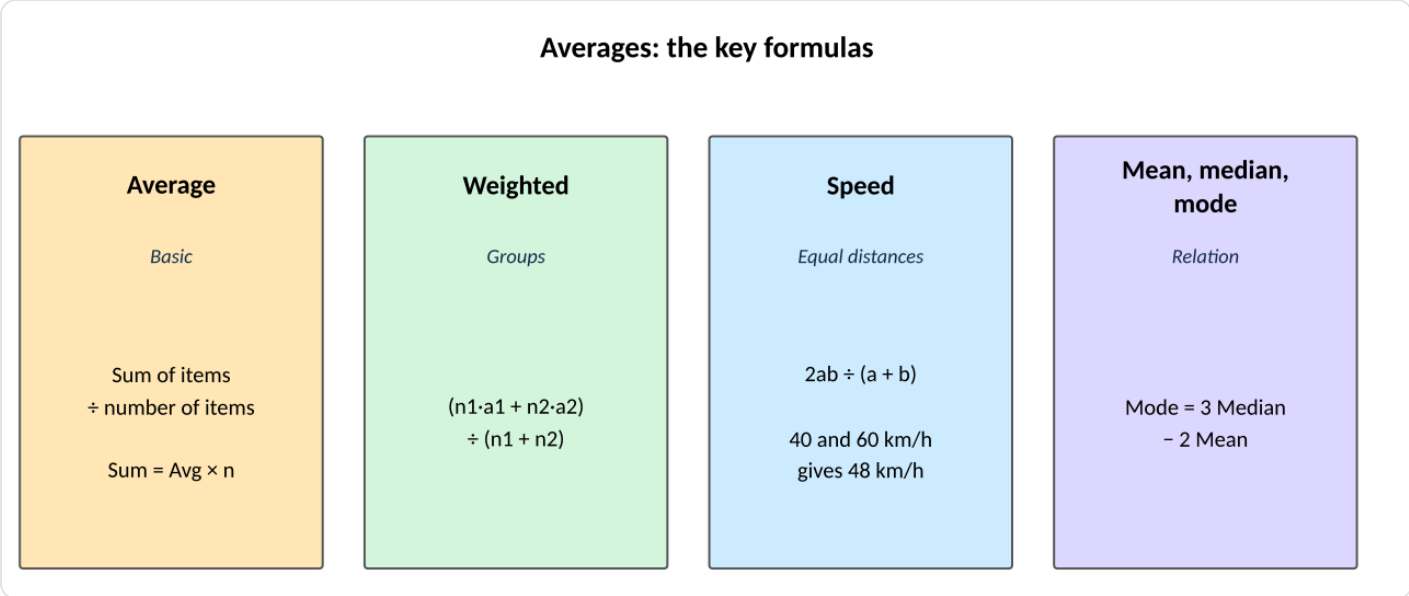
RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the President's election, qualifications, impeachment and powers. 2. Write Art 74 and Art 75 with the 42nd, 44th and 91st Amendments. 3. Draw the chart: President → PM → Council → Lok Sabha.

MATHS • Averages

UNDERSTAND IT FIRST

What is it? An **average** (arithmetic mean) is the **one number that could replace every item** and leave the **total unchanged**. If 5 students score 40, 50, 60, 70, 80, the total is 300; if each had scored the average, 60, the total would still be 300. So: **Average = Sum ÷ Number of items**, and therefore **Sum = Average × Number**. Nearly every averages problem is just moving between **sum** and **average**.



Key formulas of averages

1. Shortcuts and properties

Situation	Rule	Example
First n natural numbers	Average = **(n + 1) ÷ 2**; sum = n(n+1)/2	First 20: average 10.5
First n even / odd numbers	Even: average **n + 1**; odd: average **n**	First 10 odd numbers: 10
Any AP / consecutive numbers	Average = **(first + last) ÷ 2** = the **middle** term	7, 14, … 98: (7 + 98) ÷ 2 = 52.5
Add 1 to every item	Average rises by the same amount; **multiply** every item and the average is multiplied too	Average 20; each +5: new 25
Add a new item (n items, average A, the new average A′)	New item = **A′ × (n + 1) − A × n**	Class of 30, average 14; with teacher average 15: teacher = 15×31 − 14×30 = **45**
Remove an item	Removed item = **old sum − new sum**	5 numbers, average 27; after removal average 25: 135 − 100 = **35**
Replace an item	Change in average × n = **new item − old item**	
Combined (weighted) average	$(n_1a_1 + n_2a_2) \div (n_1 + n_2)$	40 students at 60, 60 students at 70: (2,400 + 4,200) ÷ 100 = **66**
Average speed	**Total distance ÷ total time**. For **equal distances** at a and b: **2ab ÷ (a + b)**, not (a + b) ÷ 2. For **equal times**: (a + b) ÷ 2.	40 km/h and 60 km/h over equal distances: 2×40×60 ÷ 100 = **48** km/h

EXAMPLE

Why not 50? A car goes 120 km at 40 km/h (3 hours) and 120 km at 60 km/h (2 hours). Total 240 km in 5 hours = **48 km/h**, not 50. The car spends **more time** at the slower speed, so the average sits **nearer the slower** one.

2. Mean, median and mode

- **Mean** = sum ÷ number. **Median** = the **middle value** when the data are arranged in order (for an even number of items, the **average of the two middle values**). **Mode** = the value that occurs **most often**.
- For a moderately skewed distribution: **Mode = 3 × Median − 2 × Mean**. Example: Mean 20, Median 18 gives Mode = 54 − 40 = **14**.
- **Weighted average** (marks with different weights, or a mixture): multiply each value by its weight, add, and divide by the total weight. **Deviation method:** pick a guess average, add the **differences** from it, divide by n and add back, which saves effort when numbers are close together, e.g. 48, 52, 50, 49, 51 (guess 50: differences −2, +2, 0, −1, +1 sum 0, so the average is **50**).

EXAM TRAP

Sum = Average × n; average of **1 to n** = (n + 1)/2; of an **AP** = (first + last)/2; **new item = A′(n + 1) − An**; **equal distances: 2ab/(a + b)**; **Mode = 3 Median − 2 Mean**; **combined average** = weighted by the number of items.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

The average of five numbers is 27. If one number is removed, the average of the remaining numbers is 25. What is the number removed?

A. 30

B. 35

C. 40

D. 45

Answer: B. Sum = 135; after removal sum = 4 × 25 = 100; removed number = 35.

PRACTICE

A car covers the first half of a journey at 40 km/h and the second half of the same distance at 60 km/h. What is its average speed for the whole journey?

A. 50 km/h

B. 48 km/h

C. 52 km/h

D. 45 km/h

Answer: B. 2ab/(a+b) = 2 × 40 × 60 / 100 = 48 km/h.

PRACTICE

The average age of 30 students in a class is 14 years. When the teacher's age is included, the average rises by 1 year. What is the teacher's age?

A. 40 years

B. 42 years

C. 45 years

D. 46 years

Answer: C. New total = 15 × 31 = 465; old total = 14 × 30 = 420; teacher = 45.

PRACTICE

For a moderately skewed distribution, the mean is 20 and the median is 18. What is the approximate mode?

A. 14

B. 16

C. 22

D. 24

Answer: A. Mode = 3 × Median − 2 × Mean = 54 − 40 = 14.

PRACTICE

In a coaching centre, 40 students of Batch A have an average of 60 marks and 60 students of Batch B have an average of 70 marks. What is the average of all 100 students?

A. 65

B. 66

C. 67

D. 68

Answer: B. (40 × 60 + 60 × 70) / 100 = (2,400 + 4,200) / 100 = 66.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the basic formula and 6 shortcuts. 2. Solve the speed, new-item and weighted examples without looking. 3. Write the mean–median–mode relation.

NOTEBOOK PAGE · Day 10: write this from memory

Close the notes. Write these headings, fill every blank, then correct in a different colour.

HISTORY: Medieval art, architecture, language, literature

Area	Write
Features	Arch, dome, minaret, lime mortar; calligraphy and arabesque instead of figures; jali; pietra dura; Hindu: gopuram, mandapa
Delhi Sultanate	Qutb Minar (Aibak, Iltutmish), Quwwat-ul-Islam, Adhai-din-ka-Jhonpra, Alai Darwaza 1311, Tughlaqabad, Lodi double dome
Provincial	Adina (Pandua), Atala (Jaunpur), Sidi Saiyyed jali, Mandu, Gol Gumbaz, Charminar 1591, Bidar Madrasa
Vijayanagara	Hampi: Virupaksha, Vittala and stone chariot, Hazara Rama, Lotus Mahal; Lepakshi, Tadipatri; gopurams, kalyana mandapas
Painting, music, dance	Persian, Mughal, Rajput, Pahari, Deccani; Khusrau, Tansen, Dhrupad, Purandaradasa; Kathak, Kuchipudi
Literature	Al-Biruni, Barani, Khusrau, Ibn Battuta, Kalhana, Jayasi; Telugu: Nannaya, Tikkana, Errana, Srinatha, Potana, Krishnadevaraya and Ashtadiggajas, Annamacharya

POLITY: Parliamentary executive

- President: Arts 52–62, electoral college (elected MPs + elected MLAs, STV), 35 years, 5-year term, impeachment Art 61 (2/3, 14 days notice), powers: executive, legislative, financial, judicial, military, emergency, veto (absolute, suspensive, pocket).
- PM and Council: Art 74 advice binding (42nd, 44th), Art 75: appointment, 15% cap (91st), collective responsibility to LS, individual responsibility at President's pleasure, 6 months to become a member; Cabinet, MoS (IC), MoS.

MATHS: Averages

Write: Average = Sum ÷ n; Sum = Avg × n; weighted average; average speed for equal distances = 2ab ÷ (a + b); average of consecutive numbers and an AP = (first + last) ÷ 2; effect of adding or removing an item; Mode = 3 Median − 2 Mean; solve one example of each.